Name: Mustansir Rangwala

Batch: B2

3083

Classification using Deep neural network (Any One from the following)

1. Multiclass classification using Deep Neural Networks: Example: Use the OCR letter recognition dataset https://archive.ics.uci.edu/ml/datasets/letter+recognition

2. Binary classification using Deep Neural Networks Example: Classify movie reviews into positive" reviews and "negative" reviews, just based on the text content of the reviews. Use IMDB dataset

In [ ]:
```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score
import io
```

In [ ]:
```
dataset = pd.read_csv('letter_recognition.data', sep = ",", header = None)
```

In [ ]:
```
X = dataset.iloc[:, 1 : 17]
Y = dataset.select_dtypes(include = [object])
```

In [ ]:
```
X_train, X_validation, Y_train, Y_validation = train_test_split(X, Y, test_size = 0.2
```

In [ ]:
```
scaler = StandardScaler()
scaler.fit(X_train)
```

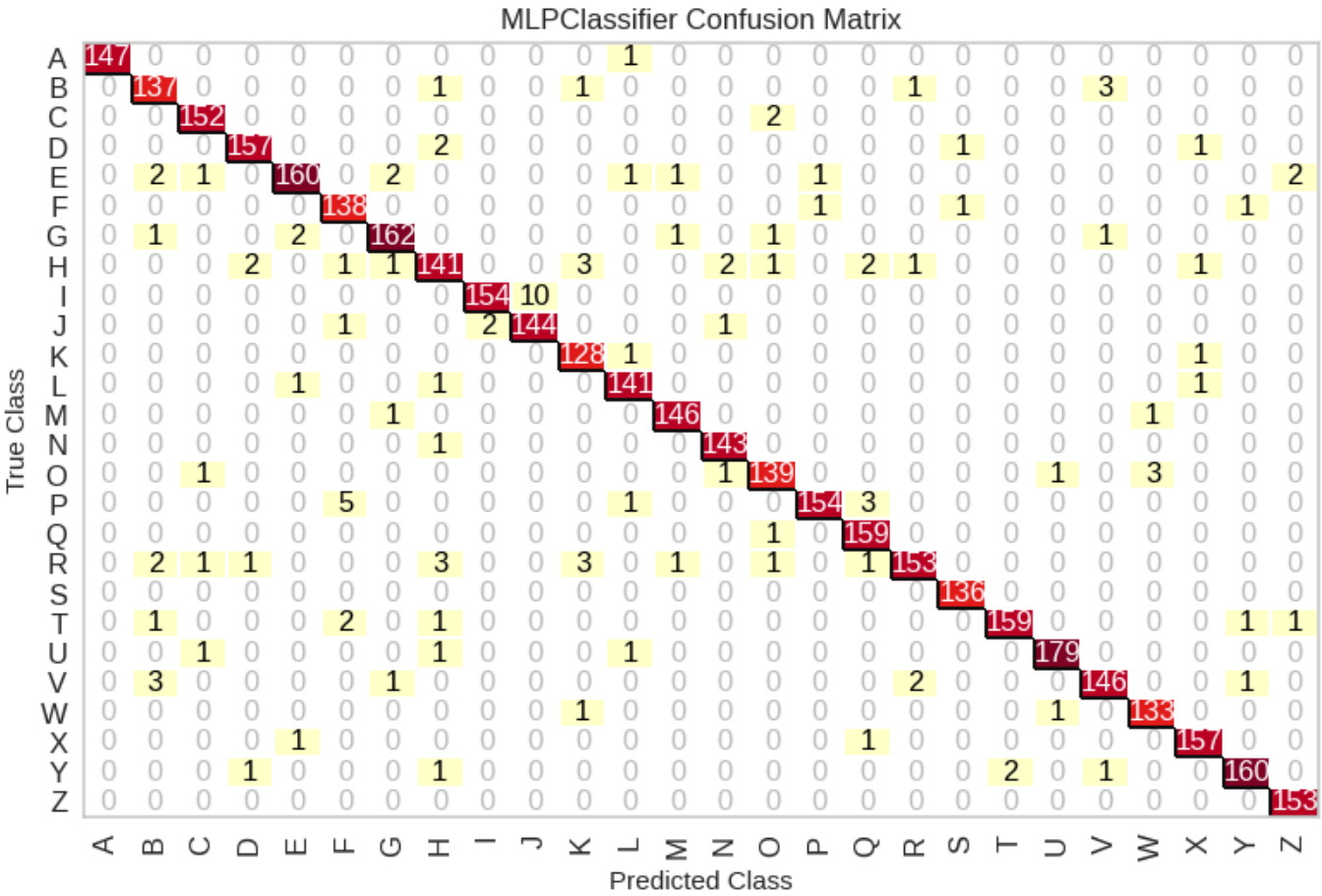
Out[ ]:
```
▾ StandardScaler
StandardScaler()
```

In [ ]:
```
X_train = scaler.transform(X_train)
X_validation = scaler.transform(X_validation)
```

In [ ]:
```
mlp = MLPClassifier(hidden_layer_sizes = (250, 300), max_iter = 1000000, activation =
```

In [ ]:
```
from yellowbrick.classifier import confusion_matrix
cm = confusion_matrix(mlp,X_train,Y_train, X_validation, Y_validation, classes="A,B,C
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/neural_network/_multilayer_perceptron.
py:1124: DataConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples, ), for example using ravel().
  y = column_or_1d(y, warn=True)
/usr/local/lib/python3.12/dist-packages/sklearn/preprocessing/_label.py:110: DataConve
rsionWarning: A column-vector y was passed when a 1d array was expected. Please change
the shape of y to (n_samples, ), for example using ravel().
  y = column_or_1d(y, warn=True)
```



In [ ]:
```
# !pip install yellowbrick
```

In [ ]:
```
cm.fit(X_train, Y_train.values.ravel())
```

Out[ ]:

ConfusionMatrix

```
ConfusionMatrix(ax=<Axes: title={'center': 'MLPClassifier Confusion Matrix'}, xlabel='Predicted Class', ylabel='True Class'>,
                classes=['A', 'B', 'C', 'D', 'E', 'F', 'G', 'H', 'I', 'J', 'K',
                         'L', 'M', 'N', 'O', 'P', 'Q', 'R', 'S', 'T', 'U', 'V',
                         'W', 'X', 'Y', 'Z'],
                cmap=<matplotlib.colors.ListedColormap object at 0x7c56432a3110>,
                estimator=MLPClassifier(activation='logistic',
                                        hidden_layer_sizes=(250, 300),
                                        max_iter=1000000))
```

estimator: MLPClassifier

```
MLPClassifier(activation='logistic', hidden_layer_sizes=(250, 300),
              max_iter=1000000)
```

MLPClassifier

```
MLPClassifier(activation='logistic', hidden_layer_sizes=(250, 300),
              max_iter=1000000)
```

In [ ]:
```
cm.score(X_validation, Y_validation)
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/preprocessing/_label.py:110: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples, ), for example using ravel().
  y = column_or_1d(y, warn=True)
```

Out[ ]: 0.9695

In [ ]:
```
predictions = cm.predict(X_validation)
predictions
```

Out[ ]: array(['C', 'F', 'O', ..., 'S', 'X', 'R'], dtype='<U1')

In [ ]:
```
print("Accuracy: ", accuracy_score(Y_validation, predictions))
```

```
Accuracy:  0.9695
```